Cardiac Deformation Lab

Quantitative Multi-Analysis Comparison & Kinematic Variance Report

Analyses Compared:	**2 Analyses**	**Generation Date:**	October 16, 2026 at 04:45:05 UTC
Study Context:	All selected analyses belong to the same study: 'Verification Study A'.	**Calibration Compatibility:**	Image-Space Only
Analysis Region Scope:	Full Image	**Primary Motion Model:**	Dense Farnebäck Optical Flow
Analysis 1 (single_pair):	60c193af-712... (2026-10-16T04:45:04)	**Scope / Spacing 1:**	Full Image • Pixels (px)
Analysis 2 (single_pair):	110bcfc1-ef7... (2026-10-16T04:45:04)	**Scope / Spacing 2:**	Full Image • Pixels (px)

1. Largest Quantitative Differences

Top computed metric variances between Analysis 1 (baseline) and Analysis 2. These differences reflect algorithmic optical-flow divergences; they do NOT indicate clinical significance.

Metric Parameter	Category	Analysis 1	Analysis 2	Difference (Δ)	% Difference
Deformation Energy Proxy	Advanced Deformation Descriptors	0.0620	0.2176	+0.1556 dimensionless	**+251.00%**
Mean Radial Strain	Ventricular Geometric Deformation	0.3288	-0.4265	-0.7552 dimensionless	**-229.70%**
Strain Gradient Magnitude	Advanced Deformation Descriptors	0.0328	0.0782	+0.0454 px■¹	**+138.29%**
Mean Displacement	Displacement Kinematics	0.9898	2.0314	+1.0415 px	**+105.22%**
Displacement Std Dev	Displacement Kinematics	1.9744	3.9350	+1.9606 px	**+99.30%**

2. Displacement Kinematics Comparison

Metric Parameter	Unit	Analysis 1	Analysis 2	Difference (Δ)	% Difference
Mean Displacement	px	0.9898	2.0314	+1.0415	+105.22%
Maximum Displacement	px	6.7575	13.0266	+6.2690	+92.77%
Minimum Displacement	px	N/A	N/A	N/A	N/A
Displacement Std Dev	px	1.9744	3.9350	+1.9606	+99.30%

3. Small-Deformation Continuum Strain Comparison

Metric Parameter	Unit	Analysis 1	Analysis 2	Difference (Δ)	% Difference
Mean Exx (Normal X)	dimensionless	N/A	N/A	N/A	N/A
Mean Eyy (Normal Y)	dimensionless	N/A	N/A	N/A	N/A
Mean Exy (Shear)	dimensionless	N/A	N/A	N/A	N/A
Mean Principal Strain (ε■)	dimensionless	N/A	N/A	N/A	N/A

Maximum Principal Strain (ε■)	dimensionless	N/A	N/A	N/A	N/A

4. Advanced Deformation Metrics Comparison

Metric Parameter	Unit	Analysis 1	Analysis 2	Difference (Δ)	% Difference
Motion Coherence	dimensionless [0-1]	0.8923	0.8761	-0.0162	-1.82%
Strain Gradient Magnitude	px■⁻¹	0.0328	0.0782	+0.0454	+138.29%
Strain Anisotropy Index	dimensionless [0-1]	0.4081	0.4324	+0.0244	+5.97%
Strain Entropy	bits	1.9812	1.1042	-0.8770	-44.26%
Deformation Energy Proxy	dimensionless	0.0620	0.2176	+0.1556	+251.00%

5. Ventricular Geometric Deformation Comparison

Metric Parameter	Unit	Analysis 1	Analysis 2	Difference (Δ)	% Difference
Mean Radial Strain	dimensionless	0.3288	-0.4265	-0.7552	-229.70%
Mean Circumferential Strain	dimensionless	0.0014	0.0025	+0.0011	+77.20%

6. Multi-Frame Temporal Trajectory Comparison

Metric Parameter	Unit	Analysis 1	Analysis 2	Difference (Δ)	% Difference
Temporal Stability Index (TSI)	dimensionless [0-1]	N/A	N/A	N/A	N/A
Temporal Mean Principal Strain	dimensionless	N/A	N/A	N/A	N/A
Temporal Strain Std Dev	dimensionless	N/A	N/A	N/A	N/A
Temporal Minimum Strain	dimensionless	N/A	N/A	N/A	N/A
Temporal Maximum Strain	dimensionless	N/A	N/A	N/A	N/A

7. Cardiac Deformation Fingerprint (10-D) Comparison

Dimension Symbol	Canonical Descriptor	Analysis 1	Analysis 2
D_mean	Mean Displacement (px)	N/A	N/A
D_max	Maximum Displacement (px)	N/A	N/A
epsilon_mean	Representative Mean Strain	N/A	N/A
epsilon_max	Representative Max Strain	N/A	N/A
epsilon_principal	Representative Principal Strain	N/A	N/A
AI	Strain Anisotropy Index	N/A	N/A
H	Strain Entropy	N/A	N/A
G_epsilon	Strain Gradient Magnitude	N/A	N/A
C	Motion Coherence	N/A	N/A
TSI	Temporal Stability Index	N/A	N/A

8. Scientific Methodology & Non-Diagnostic Protocol Disclaimer

Comparison Methodology: This comparative analysis compiles computed 2D continuum strain tensors, Farnebäck optical flow kinematics, and derivative scalar metrics calculated from pixel arrays. Differences across analyses reflect algorithmic variations in spatial gradients, regional displacement, or temporal stability.

Important Limitations: *Image acquisition frame rates, acoustic window variations, contrast-to-noise ratio, spatial resolution, and Region of Interest (ROI) demarcation directly impact motion estimation. Uncalibrated analyses express displacement solely in image pixels (px).* ***This comparison summarizes quantitative image-analysis measurements and is NOT a clinical diagnostic assessment.*** *Quantitative differences do NOT imply pathological abnormality, clinical improvement, or disease classification.*